

Lab 01: C Fundamentals

CSE 4108

Structured Programming I Lab

Lab Tasks

1. **Counting Geese:**

Mr. Poached Eggsy is the owner of a magical goose farm. A magical goose only lays **13** platinum eggs at the beginning of the year. The eggs take **6 months** to hatch and another **6 months** are required for the goslings to mature. So, the new geese can lay eggs right after one year. However, Mr. Eggsy sells half of all the eggs each year to make his ends meet. In case he has odd number of eggs, e.g. 15, he will sell 7 eggs and keep 8 eggs to increase the number of geese in his farm.

Given the number of geese `n` at the beginning of first year as input, calculate how many geese Mr. Eggsy will have after the 1st year, 2nd year and 3rd year.

Sample Output:

```
Enter initial geese count: 3
Geese count after year 1: 23
Geese count after year 2: 173
Geese count after year 3: 1298
```

[Hint] Be sure to use integers for all the values. Dividing integer by integer will always result in integer. For instance, 7/2 = 3. In these odd cases, the other half can be calculated by subtracting first half from total, i.e 7 - 3 = 4. If a problem seems difficult, first try to break it down and do it on pen and paper. Once the actual logic is clear to you, then begin coding This process, in some cases, is referred to as 'Dry Run'.

2. **Polynomial Evaluation:**

Write a C program that asks the user to enter the value of x and outputs the value of the following polynomial:

$$3x^5 + 2x^4 - 5x^3 - x^2 + 7x - 6$$

Hint: C doesn't have an exponentiation operator, so you'll need to multiply ***x*** by itself repeatedly in order to compute the power of ***x***. (For example: ***x*x*x*** is ***x*** cubed)

3. **Horner's Rule:**

Modify the program of the last task so that the polynomial is evaluated using the following formula:

$$((((3x + 2)x - 5)x - 1)x + 7)x - 6$$

Note that the modified program performs fewer multiplications. This technique for evaluating polynomials is known as *Horner's Rule*.

4. **HH:MM:SS**

You are given an integer value $seconds$, where $0 < seconds < 356400$. Your task is to convert the value into hours, minutes and seconds and print it in the format of HH:MM:SS, where HH represents the hours, MM represents the minutes and SS represents the seconds.

Sample Run 1:

```
Enter total seconds: 23
Output in required format: 00:00:23
```

Sample Run 2:

```
Enter total seconds: 4000
Output in required format: 01:06:40
```

5. **Mountain Climbing**

The Clever Climbers go mountain climbing regularly. Their mountain climbing expeditions go on for several hours and each member has a maximum height (in meters) that they can climb in an hour. In the first hour, the climber has lots of energy and climbs their maximum height

during that hour. For every subsequent hour, the climber's energy falls and they can only climb **50%** of the height they climbed on the previous hour.

The climber always climbs an integer number of meters in an hour. In the case that the climber needs to climb a fractional value, they get lazy and only climb till the nearest whole number.

Given the maximum height n that the climber can climb in an hour as input, calculate height the climber will reach after the 1st hour, 2nd hour and 3rd hour.

Sample Run:

```
Enter maximum height: 64
Height after hour 1: 64
Height after hour 2: 96
Height after hour 3: 112
```